

AIDI Capstone II

Evaluation: Final Project Report & Code and Career Self-Reflection

Weight: 30 points

Due Date: 7 August 2026

Final Project Objective:

This evaluation assesses the extent to which the AI-based product has progressed from a Minimum Viable Product (MVP) to a launch-ready solution. The student successfully identified and implemented enhancements to existing features, including improvements to algorithm performance, informed by user and stakeholder feedback. User-centered design principles were effectively applied, and essential MLOps practices—such as automation, monitoring, and model retraining—were established to support deployment readiness. Additionally, internal-facing features (e.g., analytics dashboards) were developed to enable product scalability, performance tracking, and long-term user engagement. Overall, the completed work should demonstrate a clear, well-executed transition from MVP to a scalable, real-world AI product suitable for deployment.

By completion of the project students will have become proficient in identifying business opportunities, leverage the dimensions of AI technologies, and build on the success of existing AI applications to improve products and services in any industry.

Instructions/Required:

GROUP REPORT: The final project **report** should include AT LEAST three sections: 1) The business model 2) The technical details 3) Project Management. Other sections might include research performed, technical references, market studies, financing considerations, real-world deployment issues, security and privacy concerns, maintaining system (model) integrity, run environment (AWS etc.), cost estimates for development, operations and more. This report should be GROUP effort and will result in a GROUP GRADE. See **APPENDIX A** for more details.

INDIVIDUAL REPORT: This “report” consists of a **VIDEO** submission made by each individual group member. See **APPENDIX B** for more details.

Submissions Required (Group Only):

1. Your final group report (minimum 6,000 – 11,000 words)
2. Link to all GROUP code via password-less link to Github or other
3. At least 15 strong references used in your report and listed in a bibliography, LIVE LINKED
4. All upload to DC CONNECT by the due date August 7, 2026, by one student only (not all group students)

Submissions Required (Each Individual Student):

1. 8 – 12 minute individual video, as described in Appendix B
2. Link to all INDIVIDUAL code via password-less link to Github or other
3. All upload to DC CONNECT by the due date August 7, 2026, by each individual student

Note 1: The instructor may attempt to execute portions of your code, analyze it or both.

Note 2: Groups are required to provide, no later than 10 days before the product demo, information about file formats and other testing parameters so that the Professor may STRESS TEST the demo in real time, as determined.

Note 3: Determining what is “group” code and what is “individual code” is a matter for the groups and students to determine. Generally, if one individual creates, tests, deploys and integrates a major component of the product code approximately 75%+ by themselves, this is individual code. However, if testing is done by others, integration by others, modifications by others, then this is arguably “group” code. Designation of code as “group” or “individual” is mostly for information purposes. Some individuals involved in PM or documentation may not have any code of their own. Remember, every group member gets essentially the same grade, plus or minus a few adjustments. This does NOT apply to the video portion of the work, which is 100% individual.

Descriptive Rubric (Basic) [Quantitative Rubric Follows After]:

Criteria	Excellent	Good	Fair	Poor
Business Model	The business model is clear, concise, and well researched. It includes a thorough market analysis, customer segmentation, revenue model, and competitive landscape.	The business model is clear and includes a market analysis, customer segmentation, revenue model, and competitive landscape.	The business model is somewhat clear but lacks depth in some areas. It may be missing a market analysis, customer segmentation, revenue model, or competitive landscape.	The business model is unclear and lacks key components such as a market analysis, customer segmentation, revenue model, or competitive landscape.
Technical Details	The technical details are comprehensive and well-documented. They include a detailed description of the algorithms, data preprocessing, feature engineering, evaluation metrics, optimization techniques, deployment process, and ethical considerations.	The technical details are clear and well-documented. They include a description of the algorithms, data preprocessing, feature engineering, evaluation metrics, optimization techniques, deployment process, and ethical considerations.	The technical details are somewhat clear but lack depth in some areas. They may be missing a description of the algorithms, data preprocessing, feature engineering, evaluation metrics, optimization techniques, deployment process, or ethical considerations.	The technical details are unclear and lack key components such as a description of the algorithms, data preprocessing, feature engineering, evaluation metrics, optimization techniques, deployment process, or ethical considerations.

Project Management	The project management section is comprehensive and well-documented. It includes a description of the project objectives and scope, project team, project timeline, budget and resources, risk management, communication and reporting, and project evaluation.	The project management section is clear and well-documented. It includes a description of the project objectives and scope, project team, project timeline, budget and resources, risk management, communication and reporting, and project evaluation.	The project management section is somewhat clear but lacks depth in some areas. It may be missing a description of the project objectives and scope, project team, project timeline, budget and resources, risk management, communication and reporting, or project evaluation.	The project management section is unclear and lacks key components such as a description of the project objectives and scope, project team, project timeline, budget and resources, risk management, communication and reporting, or project evaluation.
Individual Contribution	Consistently engages with the team, actively participates in discussions, and contributes valuable insights. Contributed significantly and consistently, taking initiative to advance the project. Demonstrated excellent collaboration skills, supports team members, and resolves conflicts constructively.	Frequently participates but may not always engage proactively. Contributes satisfactorily, performing assigned tasks without taking extra initiative. Generally, collaborates well but may have occasional lapses in teamwork.	Occasionally participates but often remains passive. Contributes minimally, doing only what is asked at the basic level. Shows limited collaboration, sometimes working in isolation or not supporting team efforts.	Rarely participates and shows minimal engagement. Contributes very little or not at all to the project work. Does not collaborate, causing friction or hindering the team's progress.

Instructor Engagement & Feedback Integration	Consistently provides proactive and well-documented updates to the instructor, actively seeks and effectively incorporates feedback, and demonstrates clear iteration. Final deliverables strongly reflect continuous improvement and alignment with instructor guidance.	Provides regular updates with minor gaps, incorporates most feedback with reasonable improvements, and demonstrates some iteration. Final work reflects most instructor input.	Provides occasional updates with limited detail, incorporates some feedback but misses key suggestions, and shows minimal iteration. Final work is only partially aligned with instructor guidance.	Rarely provides updates, shows little to no effort in incorporating feedback, and lacks iteration. Final deliverables do not reflect instructor guidance.
Overall Quality	The report is well-written, organized, and visually appealing. It is easy to understand and provides a comprehensive overview of the project.	The report is well-written and organized. It provides a clear overview of the project, but may lack some detail in certain areas.	The report is somewhat disorganized and may be difficult to follow in some areas. It may also lack detail in several areas.	The report is poorly written and organized. It is difficult to understand and lacks detail in several areas.

Scoring Matrix (Quantitative Rubric) [Descriptive Rubric Precedes]

Assignment is worth **30 points**. Bonus points are possible. Summary and detailed matrices are below:

Business Model	9%
Technical	22%
Project Management	7%
Research & References	7%
Code	11%
Video (Individual)	16%
Ethical Issues	7%
Professional, well-thought out, mature, complete	21%
Total	100%

Adequacy and realism of business model	8.0%
Technical details, completeness	9.7%
Technical maturity, design	4.4%
Features & quality	2.7%
Testing & validation	4.4%
Ethical considerations	7.1%
Project management effectiveness	7.1%
Report is complete, professional, well thought out	8.8%
Adequacy of research & references	7.1%
Code appears robust, well designed, well tested	6.2%
Code appears suitable and optimized for application	5.3%
Length, quality of video - INDIVIDUAL	7.1%
Video content thoughtful, honest, insightful - INDIVIDUAL	8.8%
Overall	13.3%
WEIGHTED GRADE	100.0%

Project Report (Group Submission)

The following is considered the MINIMUM required for your report. You may enhance your submission by including other sections such as research performed, technical references, market studies, financing considerations, real-world deployment issues, security and privacy concerns, maintaining system (model) integrity, run environment (AWS etc.), cost estimates for development, operations and more. Reports should be uploaded to DC Connect by only ONE student in the group.

Be sure to include a PASSWORD-LESS Github link to the final code or some alternative method of accessing it. This should be code which is agreed as GROUP CODE, not individual code. NOTE: The instructor may attempt to execute portions of your code, analyze it or both.

1. The business model (900 – 1500 words)

The business model section should explain how your project is going to create value for your target audience and how you plan to generate revenue from it. This should include the following details (but not limited to):

- An overview of the project and its goals. Problem statement (motivation, scope of the project).
- Problem framework (identify various components of your project. Develop visual mind map if desired).
- Vision and mission statement. Imagine that you are a start up specialising in the product you are developing. A vision statement is a declaration of an organization's objectives, intended to guide its internal decision-making. A vision statement must state the current and future objectives of an organization. Your mission should succinctly describe the product or service, your customers, and the problem you're solving.

- The Business Model Canvas. Who are the target audience and their needs. Key partners. Value proposition. Cost structure. Revenue stream. An explanation of how the project meets the needs and creates value for the audience
- A discussion of the competitive landscape and how your project differs from existing solutions.
- Data and Technology Risk Assessment.
- A revenue model that outlines how you plan to monetize the project (e.g., subscription, advertising, etc.)
- A marketing strategy that explains how you plan to reach your target audience and grow your user base.
- A financial forecast that includes projected revenue and expenses over a set period of time.

2. The technical details (4000 – 7500 words)

The technical details section should describe how your project works and the technologies you used to build it. Here are some things you might consider including:

➤ **Product Maturity and Value Delivery**

- ✚ How did the core value proposition was validated and refined beyond the MVP stage. Improvements were appropriately prioritized based on user needs and stakeholder feedback, resulting in a focused and production-oriented product.

✚ Literature Review

➤ **Feature Enhancement and Technical Quality (Very Important)**

- ✚ Implemented meaningful enhancements to existing features, addressing limitations of the MVP. Evidence of improved performance, reliability, and reduced technical debt was demonstrated, indicating readiness for real-world use.
- ✚ An end-to-end project on the dataset selected. Explain your dataset in detail. Explain the EDA. Data Pipeline

-  Key improvements made to your product. Highlight any new features or functionality added.
 -  What are your products three AI features. All teams should have a minimum of three AI features.
 -  Explain the concept selection for model building part. Explain the concept selection for model building part. Which ML models you have tried and which one was finally selected? Why was that particular model selected? How was the final model fine-tuned and optimized? If you have chosen a neural network model, please show the architecture diagram. How was it made sure the developed model did not have any fitting problems?
- **User-Centered Design and Usability**
-  The product reflects the application of user-centered design principles. Enhancements to user experience, usability, and onboarding were clearly articulated and supported by feedback or testing results.
- **Operational and Deployment Readiness**
-  How is production-ready infrastructure, including appropriate deployment practices established. The solution demonstrates an understanding of scalability, maintainability, and operational stability.
 -  How was observability added into your system. Define your Service Level Indicators (SLIs). Establish Service Level Objectives (SLOs). How did you set up logging, monitoring, and alerting to detect failures quickly.
 -  Deployment Strategy
- **Monitoring, Analytics, and Internal Tooling**
-  Appropriate monitoring and analytics mechanisms were implemented (e.g., dashboards, metrics, logging) to track system performance and user engagement. These internal-facing features support ongoing improvement and informed decision-making.

-  Explain how your product is deployed and monitored. Discuss observability, automation, or monitoring strategies.
-  How did you create/present user analytics that can later guide future improvements and user retention strategies.
- **Testing and Validation (VERY IMPORTANT SECTION):**
 -  Discuss the testing strategies used to evaluate your product (functional testing, model performance, system testing, etc.). Share any metrics or results that demonstrate product improvement.
- **AI/MLOps Practices**
 -  For AI-based products, how is effective model management practices demonstrated, including performance evaluation, monitoring, and readiness for retraining, reflecting an understanding of real-world AI deployment challenges.
- **Ethical considerations (Very Important).**
 -  If your project involves using AI to make decisions that impact people's lives, you should describe any ethical considerations you took into account. Explain how you addressed issues such as bias, fairness, and transparency in the model's decision-making process.
 -  How did you ensure security and compliance.

3. Project Management (1100 – 2000 words)

- ✓ What is the project management philosophy used? (Eg: agile scrum)
- ✓ Project team: Describe the members of the project team, including their roles and responsibilities. This section should outline the team's structure, communication channels, and workflow processes.

- ✓ Project timeline: Create a timeline that outlines the major milestones and deliverables of the project. This section should include a detailed schedule that identifies the start and end dates of each task, as well as any dependencies or critical paths that could impact the project's timeline.
- ✓ Everyone's role, method of communication
- ✓ Team and individual review of each other's work
- ✓ Issues, difficulties, workload balancing, personality issues and how resolved
- ✓ Communication and reporting: Describe the project's communication plan, including how progress was reported and how stakeholders will be kept informed. This section should outline the project's reporting schedule, including the frequency, format, and content of progress reports.
- ✓ Budget and resources: Describe the project budget and the resources used to complete the project. This section should include a breakdown of the project costs, including labor, materials, and equipment.

4. Other Requirements

- What is the Collaborative platform + version control software you used? Why did you use this? Alternatives considered?
- References: Include **AT LEAST 15 STRONG, RELEVANT, LIVE LINKED REFERENCES** related to your product, its market, the technology, the financial considerations, AI security and privacy, legal requirements relevant to your project, deployment and environment issues. Include these in a bibliography at the end of your report. **EACH REFERENCE MUST BE CITED AT LEAST ONCE IN YOUR REPORT**, via a footnote.
- **A reminder: Be sure to include a PASSWORD-LESS Github link to the final code or some alternative method of accessing it. This should be code which is agreed as GROUP CODE, not individual code.**

NOTE: The instructor may attempt to execute portions of your code, analyze it or both.

Appendix “B”

Each individual student will **record an 8 – 12 minute video** and upload to the assignment portal in DC Connect. Each individual student will also provide a link (github or other) with NO PASSWORD to software which they personally developed and which is considered “theirs”. **Individual software and group software cannot be the same (either one or the other).** **NOTE: The instructor may attempt to execute portions of your code, analyze it or both.**

Think of the video as a private chat with the Professor. It should include at minimum:

- History of your involvement with this project and this group
- Your role or roles in the group. Did they evolve from Capstone I (how, why).
- Your ideas about the future for your project – dead end or real commercial opportunities? Why? Any plans going forward?
- Group dynamics (this video is confidential) – how the group functioned (or didn’t function). Work loads, effort you expended... How could the situation have been improved.
- Do you agree with how the project/product has been developed or would you have preferred another way, another technology, different functionality? Why? Did you discuss these within your group?
- What you have learned from this project about business, technology, development, project management, research?
- Best memories and outcomes?
- Worst memories and outcomes?
- Anything else you would care to share?

END